MTH1001-A

UNIVERSITY OF EXETER

FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

MATHEMATICS AND STATISTICS

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Mathematical Structures

Module Leader: Prof. Nigel Byott

Other examiner: Dr Gihan Marasingha

Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

No Word Count specified

SECTION A

1. (a) (i) Give the complete parse tree for the expression $5 < 10 + 2 \times 3$.

   (ii) State the substitution principle.

   (iii) Let a and b be real numbers, let $f : \mathbb{R} \to \mathbb{R}$, and let P be the predicate given by $P(x) \coloneqq 2 + 5f(x) \neq xf(3b)^2$. Suppose $h_1 : 3b = a^2$ and $h_2 : 2 + 5f(3b) \neq 3bf(3b)^2$. What is the result of substituting h_1 into h_2 using P? (10)

   (b) In this question, select **one or more** possible deductions from a list of premises. You need not provide reasoning.

	Premise(s)	Deductions
(i)	p, $q \to p$	p, q, $p \wedge q$, $p \vee q$, $\neg p$, $p \to q$
(ii)	p, $q \vee r$	p, q, r, $p \vee q$, $r \vee q$, $q \vee r$
(iii)	$p \vee \neg q$	p, $p \vee q$, $p \to q$, $q \to p$, $\neg p$, $\neg q$
(iv)	$p \leftrightarrow q$, $\neg q$	p, q, $p \wedge q$, $p \vee q$, $\neg p$, $\neg q$

   (8)

   (c) (i) Let S and T be types. Let $g : S \to T$ be a function. What does it mean for g to be surjective?

   (ii) Let $p : \mathbb{Z} \to \mathbb{Z}$ and $q : \mathbb{Z} \to \mathbb{N}$ be functions where

$$p(x) \coloneqq \begin{cases} x + 10, & \text{if } x \geq -10 \\ -x, & \text{otherwise,} \end{cases} \quad \text{and} \quad q(x) \coloneqq p(x).$$

   Is p surjective? Is q surjective? Justify your answers. (10)

   (d) Consider the sets $A \coloneqq \{2, 3, 8\}$, $B \coloneqq \{2, 6\}$, $C \coloneqq \{1, 3, 5\}$, and $D \coloneqq \{\varnothing, \{2\}, \{2, 6\}, \{3, 5\}, \{1\}\}$. Write each of the following sets in finite set notation **and** state the number of elements in each set. You need not provide reasoning.

   (i) $C \cup A$, (ii) $B \cap A$, (iii) $C \setminus A$,
   (iv) $C \times B$, (v) $D \setminus \mathcal{P}(A)$, (vi) $\mathcal{P}(A) \setminus \mathcal{P}(B)$.

   (12)

   (e) Let $(c_n)_{n=0}^{\infty}$ be a sequence of real numbers.

   (i) Define what it means for $\sum_{n=0}^{\infty} c_n$ to converge.

   (ii) Prove that if $\sum_{n=0}^{\infty} c_n$ converges, then $(c_n)_{n=0}^{\infty}$ converges to 0. State clearly any result you use concerning sequences. (10)

[**50**]

SECTION B

2. (a) Give a Fitch-style proof of $(p \wedge (q \vee \neg p)) \rightarrow q$, for propositions p and q. Your proof should use only the rules of inference. (12)

   (b) Let S be a set of real numbers and let x be a real number.

   (i) Give the definition of an *upper bound* of S.

   (ii) What does it mean for x to be a *supremum* of S?

   (iii) Let A and B be non-empty sets of real numbers. Suppose that A has a supremum. Let C be the set $\{a - b^2 \mid a \in A, b \in B\}$. Prove that C has a supremum.

   (iv) With A, B, and C as in 2(b)(iii), prove that $\sup(C) \leq \sup(A)$. (13)

   [25]

3. (a) Let X, Y be types. Let $g : X \rightarrow Y$ be a function. Let P be a set of elements of X and let Q be a set of elements of Y.

   (i) Give the definition of the *pre-image* $g^{-1}(Q)$.

   (ii) Give the definition of the *image* $g(P)$.

   (iii) Let A and B be types. Let T be a set of elements of A, let $f : A \rightarrow B$ be a function and let $U \subseteq f(T)$. Prove that there exists a set S of elements of A such that $f(S) = U$. (15)

   (b) Prove, by induction, that there exists m, a natural number (a non-negative integer), such that $3^{n+m} \leq (n + m)!$ for all natural numbers n. You may use all familiar rules of arithmetic. (10)

   [25]

4. (a) In this question, A and B are non-empty sets.

   (i) Let $f : A \to B$ be an injective function. Prove that there is a subset C of B and a function $g : A \to C$ such that g is bijective and such that $g(a) = f(a)$ for every a in A.

   (ii) Let $p : A \to B$ be a surjective function. Prove that there is a subset D of A and a function $q : D \to B$ such that q is bijective and such that $q(d) = p(d)$, for every d in D. (13)

   (b) Let $(b_n)_{n=0}^{\infty}$ be a sequence of real numbers that converges to the real number b. Without using any results from the 'algebra of limits', prove that $(b_n^2)_{n=0}^{\infty}$ converges to b^2. State clearly any results about convergent sequences that you use in your proof. (12)

[**25**]